

Aaron Zoll

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EDUCATION

Johns Hopkins, Graduate

August 2023 – May 2027 (expected)

Ph.D in Applied Mathematics, Advisor: Ben Grimmer

Johns Hopkins, Graduate

August 2022 – May 2023

M.S in Applied Mathematics

Johns Hopkins, Undergraduate

August 2019 – May 2022

B.S in Applied Mathematics, B.A in Mathematics

TEACHING EXPERIENCE

Johns Hopkins — Course Instructor

Summer 2022 – Present

Johns Hopkins — Course Developer

Summer 2021

Johns Hopkins — Teaching Assistant

January 2020 – Present

Johns Hopkins — Directed Reading Program Mentor

Spring 2025/Fall 2025

Johns Hopkins — MSE Orientation Review Session

August 2023/2024

Mathnasium — Math Instructor

August 2019 – December 2019

Personal Tutor

2022 – Present

INSTRUCTOR OF RECORD

Developed complete syllabi, lecture notes, discussion pages, and assignments to facilitate courses ranging from 15 to 140 students. Presented lectures, from asynchronous online to fully in-person, on linear algebra, probability, statistics, data analysis, and differential equations.

Data Analysis Workshop (AS.110.100)

Summer 2023–2026

Johns Hopkins University

Linear Algebra and Differential Equations (EN.553.291)

Fall 2026

Johns Hopkins University

COURSE DEVELOPMENT

Contributed to the design and development of new mathematics and engineering courses. Authored original lecture material, guided Excel practice problems, engaging problem sets, and instructional resources tailored to enhance pedagogical clarity and student engagement.

College Algebra (AS.110.102)

Summer 2021

Johns Hopkins University

Data Analysis Workshop (AS.110.100)

Summer 2022

Johns Hopkins University

TEACHING ASSISTANT

Supported instruction in twenty undergraduate and graduate-level courses through grading, writing lecture notes, designing assignments, leading weekly discussion sections, and providing academic support to students. Collaborated with faculty to reinforce core course concepts and foster a strong learning environment.

Graduate Teaching Assistant

Introduction to Computational Mathematics (EN.553.385)	Spring 2025, Spring 2026
Machine Learning 1 (EN.553.740)	Fall 2025
Introduction to Convexity (EN.553.665)	Fall 2024, Spring 2024
Matrix Analysis and Linear Algebra (EN.553.792)	Fall 2023
Mathematical Game Theory (EN.553.653)	Spring 2023
Mathematical Modeling and Consulting (EN.553.400)	Spring 2023
Optimization in Finance (EN.553.661)	Fall 2022

Undergraduate Teaching Assistant

Real Analysis I (EN.553.405)	Spring 2022, Summer 2022
Cryptology and Coding (EN.553.371)	Spring 2022
Calculus II (For Biological and Social Science) (AS.110.107)	Spring 2022
Honors Discrete Mathematics (EN.553.172)	Fall 2021
Differential Equations and Applications (AS.110.302)	Fall 2021, Spring 2021
Discrete Mathematics (EN.553.171)	Spring 2021, Fall 2020
Calculus III (AS.110.202)	Fall 2020
Introduction to Computing (AS.205.205)	Spring 2020

DIRECT READING PROGRAM MENTEES AND PRESENTATIONS

Paired with up to three undergraduate students, mentoring a self-contained course on gradient descent, its convergence theory, and its guarantees. Wrote a series of lecture notes and exercises for students who were, in addition to weekly hour one-on-one sessions, expected to prepare a presentation to conclude the semester.

Lingxi Kong Fall, 2025

Performance Estimation for Smooth Convex Gradient Descent

Darius Kim Fall, 2025

Adaptive Momentum Methods for Stochastic Optimization

Hermoine Larkin Spring, 2025

ADAM applications in Distillation Procedures

Hamza Alvi Spring, 2025

Acceleration Methods for Convex Minimization

Wynn Zhao Spring, 2025

Extension of Gradient Methods for Nonsmooth Optimization

SERVICE

Johns Hopkins AMS Department External Review

Fall 2023

Ph.D. Student Committee Member

Performed review of the structure and design of the graduate program in the applied math and statistics department. Participated in discussion about course structure, qualifying exams, and various requirements for incoming and current Ph.D. students.

Johns Hopkins Education and Artificial Intelligence Focus Group

May 2026–Present

Graduate Researcher

Participated in a group of faculty, post-doctorates, graduate students, and undergraduate student workers to research, develop, and implement AI into pedagogy. With the goal to enhance student learning, I beta-tested various course-specific AI tools, provided feedback, and integrated into various Johns Hopkins courses' LMS. Further work included providing feedback on textbook manuscripts about the use, benefits and fallback, of AI for students, instructors, and researchers.

RESEARCH INTERESTS

Optimization Research

Ongoing

Dissertation research with Dr. Benjamin Grimmer, Johns Hopkins University

Conducting theoretical research on algorithm design and analysis to unify the regimes between smooth and nonsmooth convex problem classes (e.g. function exhibiting Hölder smoothness or uniform convexity). Prior work focused on heterogeneously smooth and convex compositions, calculus results expanding and characterizing dual notations between Hölder smoothness and uniform convexity, interpolation theory for inexact smooth convex functions, performance estimation over respective problem classes, and universal algorithm design. Future work entails characterizing the class of minimax optimal methods for convex Lipschitz minimization.

AI Pedagogy Research

Ongoing

Extracurricular research led by Dr. Sergey Kushnarev, Johns Hopkins University

Experimentation and implementation of various AI derived course assistants. Herein, we discuss, analyze, and test different methods for embedding the rising LLM tools into the education system. By supplying agents with structured prompts, focused on providing students with motivation and step-by-step guidance instead of direct answers, we aim to improve comprehension and intellectual capability.

Signal Processing Research

Upcoming

Planned collaboration with Dr. Mario Michelli & Kaleigh Rudge, Johns Hopkins University

Preparing to investigate spectral properties of the Discrete Fourier Transform and its connections to signal representation and harmonic analysis. Further investigation will include advancing understanding of the Fractional Fourier Transform, smoothly interpolating between the signal and frequency domains.

PUBLICATIONS

Accepted Papers

A Universally Optimal Primal-Dual Method for Minimizing Heterogeneous Compositions

IMA Journal of Numerical Analysis, 2025

Aaron Zoll, Benjamin Grimmer

arXiv:2503.07566

Preprints

A Complete Characterization of Optimal Subgradient Methods for Lipschitz Minimization

2026

Aaron Zoll, Benjamin Grimmer

arXiv:2607.19240

Inexactly Smooth Performance Estimation and New Optimized Gradient Methods

2026

Aaron Zoll, Benjamin Grimmer

arXiv:2606.01505

TALKS

On Interpolation Theory

Applied Mathematics and Statistics Student Seminar, *Johns Hopkins*

Mar, 2026

Algebra and Number Theory Guest Lecture, Clayton High School

Apr, 2026

On Inexactly Smooth Performance Estimation

Applied Mathematics and Statistics Student Seminar, *Johns Hopkins*

Sept, 2025

INFORMS Annual Meeting, *Atlanta, Georgia*

Oct, 2025

Junior MINDS Seminar, *Johns Hopkins Mathematical Institute for Data Science*

Nov, 2025

Modeling and Optimization: Theory and Applications (MOPTA), *Lehigh University*

Aug, 2026

On Minimizing Heterogeneous Compositions

Applied Mathematics and Statistics Student Seminar, *Johns Hopkins*

Mar, 2025

Junior MINDS Seminar, *Johns Hopkins Mathematical Institute for Data Science*

Apr, 2025

International Conference on Continuous Optimization (ICCOPT), *Los Angeles, California*

Aug, 2025

AWARDS

Joel Dean Award for Excellence in Teaching

2019–2020

Department of Applied Math and Statistics

Joel Dean Award for Excellence in Teaching

2019–2020

Department of Mathematics

Gordon L. and Beatrice C. Bowles Fellowship

2022–2023

Johns Hopkins University

Promotion to Teaching Fellow

2025

Department of Applied Mathematics and Statistics, Johns Hopkins University

SOFTWARE

Matlab · Python · Julia · LaTeX · Desmos